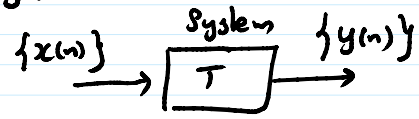
WEEK 3

29 January 2024 10:57

Linear Systems



A linear system must satisfy two conditions

1) Additivity

If inputs $\{x_1(n)\} \xrightarrow{T} \{y_1(n)\}$ outputs

$\{x_2(n)\} \xrightarrow{T} \{y_2(n)\}$

then input $\{x_1(n)\}+\{x_2(n)\}$ must produce output $\{y_1(n)\}+\{y_2(n)\}$

for any $\{x_1(n)\}$ & $\{x_2(n)\}$.

2. Homogeneity

If $\{x(n)\} \xrightarrow{T} \{y(n)\}$

then input $\alpha\{x(n)\}$ must produce output $\alpha\{y(n)\}$

(α: scalar multiple)

for any α, $\{x(n)\}$

Additivity and homogeneity are different conditions.

Example 1) $y(n) = \text{Real}\{x(n)\}$

Satisfies additivity

$$\text{Re}\{a+b\} = \text{Re}\{a\} + \text{Re}\{b\}$$

Take $\alpha = j$

$$j\,x(n) \longrightarrow \text{Re}\{j\,x(n)\}$$

$$\downarrow$$

$$-\text{Im}\{x(n)\}$$

Violates homogeneity.

Example 2:

$$y(n) = \begin{cases} \dfrac{x^2(n-1)}{x(n)} & ; \text{ if } x(n) \neq 0 \\ 0 & ; \text{ if } x(n) = 0 \end{cases}$$

Easily $\alpha\{x(n)\} \longrightarrow \alpha\{y(n)\}$

$$\frac{\alpha^2 x^2(n-1)}{\alpha\, x(n)} = \alpha\, y(n).$$

Satisfies homogeneity

Violates additivity $(a+b)^2 \neq a^2+b^2$

We can group these two conditions & have one style condition

Principle of Superposition

If inputs $\{x_1(n)\} \xrightarrow{T} \{y_1(n)\}$

$\{x_2(n)\} \xrightarrow{T} \{y_2(n)\}$

If inputs $\{x_1(n)\} \longrightarrow \{y_1(n)\}$

$\{x_2(n)\} \xrightarrow{T} \{y_2(n)\}$

then input $a\{x_1(n)\} + \{x_2(n)\}$ must

produce output $a\{y_1(n)\} + \{y_2(n)\}$

Examples of Linear Systems

- Time shift, Time reversal,
  Accumulator, Moving average.

Non linear system

$$y(n) = x^2(n)$$

x ——————————— y

6. Time Invariance Property.

If input $\{x(n)\} \xrightarrow{T} \{y(n)\}$

Time Invariant System. then input $\{x(n-N_0)\} \xrightarrow{T}$ must produce $\{y(n-N_0)\}$

for any input $\{x(n)\}$ & any N_0 shift

Example: Accumulator, Moving Avg.

$\{x(n)\} \xrightarrow{T} \{y(n)\}$

$$y(n) = \sum^{n} x(k)$$

$$y(n) = \sum_{k=-\infty}^{n} x(k)$$

Let $\{x_1(n)\} = \{x(n-N_0)\}$

$$\{x_1(n)\} \xrightarrow{T} \{y_1(n)\}$$

$$y_1(n) = \sum_{k=-\infty}^{n} x_1(k)$$

$$= \sum_{k=-\infty}^{n} x(k-N_0)$$

$l = k - N_0$

$$= \sum_{l=-\infty}^{n-N_0} x(l)$$

$$\boxed{y_1(n) = y(n-N_0)}$$

Time Varying System

① $y(n) = n\, x(n)$

$\{x_1(n)\} \rightarrow \{y_1(n)\}$

$x_1(n) = x(n-N_0)$

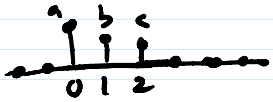
$y_1(n) = n\, x_1(n)$

$= n\, x(n-N_0)$

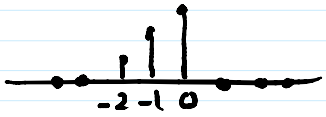
$y_1(n) \neq y(n-N_0) = (n-N_0)\, x(n-N_0)$

(2) $y(n) = x(-n)$ [Time Reversal]

$\{x(n)\} = \{\underset{\downarrow}{a}, b, c\}$

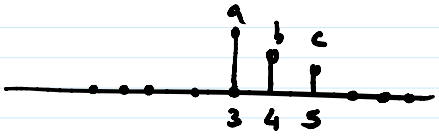


$\{y(n)\} = \{c, b, \underset{\downarrow}{a}\}$ $n=0$



$\{x_1(n)\} = \{x(n-3)\}$

$= \{\underset{\downarrow}{0}, 0, 0, a, b, c\}$

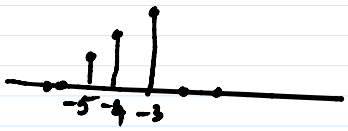


Time Reverse

$y_1(n) = x_1(-n)$

$= \{c, b, a, 0, 0, \underset{\downarrow}{0}\}$ $y_1(n)$

$y_1(n)$ =

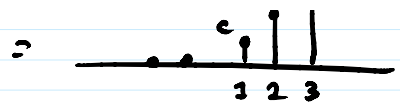


Time shift original output

$\{y(n-3)\} = \{\underset{\downarrow}{0}, c, b, a\}$

=





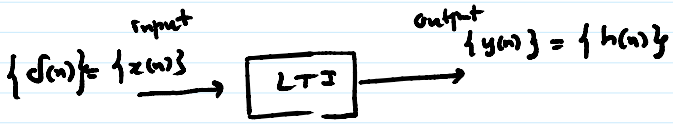


Linear Time Invariant (LTI) Systems

LTI ⇔ Linear & Time Invariant

Unit Impulse Response :

Output of DT System when input is $\{\delta(n)\}$



$\{h(n)\}$ is unit impulse response of system

An LTI System is completely characterized by its unit impulse response.

For any input $\{x(n)\}$, the LTI System output can be got by convoluing input with impulse response

Derivation of Convolution:

1. First break down input signal as weighted sum of time shifted impulse signal

2. Apply linearity & time invariance properties to obtain Convolution

1. Break down input signal

$$\{x(n)\} = \{a, b, \underset{\downarrow}{c}, d, e\}_{n=0}$$

1. Break down ...

$$\{x(n)\} = \{a, b, c, d, e\}$$ ($n=0$ at c)

$\delta(n)$ represents the entire signal / sequence

$$x(-2)\,\delta(n+2) = a\,\delta(n+2) = \{a, 0, 0, 0, 0\}$$

$$+$$

$$x(-1)\,\delta(n+1) = b\,\delta(n+1) \quad \{0, b, 0, 0, 0\}$$

$$+$$

$$x(0)\,\delta(n) = c\,\delta(n) \quad \{0, 0, c, 0, 0\}$$

$$+$$

$$x(1)\,\delta(n-1) = d\,\delta(n-1) \quad \{0, 0, 0, d, 0\}$$

$$+$$

$$x(2)\,\delta(n-2) = e\,\delta(n-2) \quad \{0, 0, 0, 0, e\}$$

Generalize this:

entire DT signal

$$\{x(n)\} = \sum_{k=-\infty}^{\infty} x(k)\,\delta(n-k)$$

$x(k)$: sample value of $\{x(n)\}$ at $n=k$

2. Convolution:

Red indicates entire signal

$T \Rightarrow$ LTI System

$$\delta(n) \xrightarrow{T} h(n)$$

$$\delta(n-k) \xrightarrow{T} h(n-k)$$

$$x(k)\,\delta(n-k) \xrightarrow[\text{homogeneity}]{T} x(k)\,h(n-k)$$

($x(k)$: scalar)

$$\sum_{k=-\infty}^{\infty} x(k)\,\delta(n-k) \xrightarrow{T} \sum_{k=-\infty}^{\infty} x(k)\,h(n-k)$$

$$\Downarrow$$

$$\{y(n)\}$$

output.

$$\Big\| \quad \{x(n)\} \text{ input}$$

$\{x(n)\}$ input — $\{y(n)\}$ output.

Convolution sum:

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)\, h(n-k)$$

Shorthand Notation: $\{y(n)\} = \{x(n)\} * \{h(n)\}$ (convolution)

Examples of LTI System.

1) Identity System

$y(n) = x(n)$ for all n.

Impulse Response: $\{h(n)\} = \{\delta(n)\}$

$$\{y(n)\} = \{x(n)\} * \{h(n)\}$$

identity:

$$\{x(n)\} = \{x(n)\} * \{\delta(n)\}$$

2. Time Shifting System

$y(n) = x(n - N_0)$; for all n

N_0 is a fixed integer.

$\{x(n)\}$ input, $\{y(n)\}$ output.

$\{x(n)\} = \{\delta(n)\}$

then $\{y(n)\} = \{\delta(n-n_0)\}$

Impulse resp.

$$\{h(n)\} = \{\delta(n - N_0)\}$$

$$\{y(n)\} = \{x(n)\} * \{\delta(n - N_0)\}$$

time shift:

$$\{x(n-N_0)\} = \{x(n)\} * \{\delta(n-N_0)\}$$